

Politecnico di Torino
Dipartimento di Automatica e Informatica
Deep NLP (01VIXSM)
Written Exam Simulation

January 7, 2026

Name: _____

Surname: _____

Student ID: _____

Exam rules:

- The present exam consists of 6 pages (including this cover page) and 7 questions overall. Any inconsistencies/printing errors in the written exam content must be reported to the teacher *at the beginning* of the exam.
- Exam duration: *60 minutes*.
- Withdraw is allowed only *at the end* of the exam.
- The exam is *closed-book*. Electronic devices, mobile phones, smart watches, and extra papers (even blank papers) are *not allowed*.
- Closed-ended questions: cross the right answer (just one) at pag. 2. Wrong or missing answers to closed-ended questions will receive *no penalty*.
- Open questions: write your answers below the text of the question. If you need more space please use the last page (i.e., pag. 6) and/or the back side of the paper.

Evaluation grid

Question:	1	2	3	4	5	6	7	Total
Points:	1	1	1	1	5	6	7	22
Score:								

1. (1 point) Given the sequence length n , the number of dimensions n , and the precomputed matrices Q , K , and V , the complexity per layer of the self-attention is
 - $O(n^2)$
 - $O(n^2d^2)$
 - $O(nd)$
 - $O(n^2d)$
 - $O(d^2)$.
2. (1 point) *Positional embeddings* in Transformers rely on
 - sinusoid and cosinusoid functions.
 - softmax functions.
 - negative exponential functions.
 - linear functions.
 - None of the above.
3. (1 point) Skip-Gram vs. CBOW
 - CBOW is slower than Skip-Gram.
 - CBOW captures semantic relations better than Skip-Gram.
 - CBOW is less sensitive to word frequency imbalances.
 - Their complexity is independent of the context size.
 - None of the above.
4. (1 point) In the context of closed-ended question answering using sequence-to-sequence models
 - The decoding stage is necessary to produce the answer.
 - The decoding stage is optional.
 - The encoding stage is optional.
 - The decoder input is undefined.
 - None of the above.

5. (5 points) Describe the main performance metrics used to evaluate text summarization systems. For each metric provide
1. Metric name.
 2. Mathematical formulation and typical value ranges.
 3. Intuition and main applications.

6. (6 points) Describe the *LSA* process.
1. Define main objective, inputs, outputs, and main parameters.
 2. Enumerate the main steps.
 3. Provide examples of practical applications (at least two).

7. (7 points) An insurance company wants to design a RAG system to handle multi-page textual documents in PDF format. Assume that the documents are written in English and averagely consist of 100+ pages. Address the following points.
1. Draw a sketch of the RAG architecture. Define inputs, outputs, and goal of each architecture component.
 2. Describe the procedures needed to evaluate RAG performance under multiple aspects with the involvement of human experts, assuming that no ground truth is given a priori.
 3. Explain how to extend the system to efficiently handle multilingual and multi-page PDF documents.

This page is intentionally left blank to accommodate work that wouldn't fit elsewhere and/or scratch work.